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Course: Calculus Beta

Assignment: MyLab Math - Uge 4

Graph the points $z = x + iy$ that satisfy the given conditions.

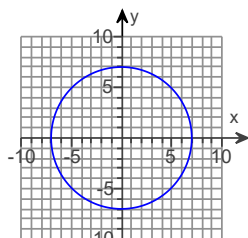
- a. $|z| = 7$ b. $|z| < 7$ c. $|z| > 7$

a. The absolute value (or magnitude) of a complex number $z = x + iy$, denoted $|z|$, is defined to be the length of a vector from the origin to the point (x, y) .

This means the points $z = x + iy$ that satisfy $|z| = 7$ are the points whose distance from the origin is 7 units.

The set of points whose distance from the origin is 7 units forms a circle of radius 7 centered at the origin.

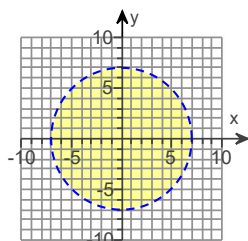
This set is shown here.



b. The points that satisfy $|z| < 7$ are the points whose distance from the origin is less than 7 units.

The set of points whose distance from the origin is less than 7 units consists of all points inside (but not on) a circle of radius 7 centered at the origin.

This set is shown here.



c. The points that satisfy $|z| > 7$ are the points whose distance from the origin is greater than 7 units.

The set of points whose distance from the origin is greater than 7 units consists of all points outside (but not on) a circle of radius 7 centered at the origin.

This set is shown here.

